

Natnael Messay

Software Developer

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Profile

Performance driven Full Stack Developer with a passion for building scalable, efficient, and impactful applications including embedded and AI-powered systems. Proficient in Data Structures and Algorithms for optimized, high performing code. Detail-oriented and committed to clean, maintainable, and well-documented practices. Experienced in creating responsive user interfaces, robust backends, and low level embedded solutions. Collaborative, adaptable, and passionate about delivering reliable, user focused software.

Education

Bachelor of Science in Computer Engineering, Addis Ababa Universty 01/2021 – 06/2026
5kilo, Addis Ababa, Ethiopia

- CGPA of 3.74
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Database Systems, Object-Oriented Programming, Computer Security, and System Architecture.
- I have solved and collaborated on 200+ algorithmic problem-solving challenges on LeetCode, strengthening analytical thinking and coding proficiency.

Skills

Front-End Html, CSS, Tailwind CSS, Javascript, React.js, Next.js	Database MYSQL, MongoDB, PostgreSQL	Version Control Git, Github
Back-End Node.js, express.js, Nest.js, Go, FastAPI	Deployment Vercel, Render, Heroku, Netlify	Machine Learning & AI Embedding AI, Supervised & Unsupervised Learning, AI APIs

Projects

News Recommendation Website 🔗

- Personalized News Recommender: Developed a full-stack application using FastAPI and React that fetches and stores news articles in PostgreSQL, delivering personalized content based on user interests.
- NLP-Driven Logic: Leveraged spaCy for Named Entity Recognition (NER) and keyword extraction to transform unstructured article descriptions into searchable metadata.
- Modern Architecture: Engineered a modular system featuring a RESTful API and a Tailwind CSS frontend, utilizing vector-based keyword matching to rank and recommend the top 5 most relevant articles.

Smart Vision Assistant for the Visually Impaired 🔗

- Built a real-time assistive navigation system on Raspberry Pi 3B using a custom-trained YOLOv8 model for object detection, spatial awareness, and instant voice feedback.
- Optimized inference performance through lightweight model tuning, dataset augmentation, and efficient embedded processing for real-time detection under varied lighting conditions.
- Integrated multimodal features including OCR-based text reading, tactile path tracking, and modular GPIO architecture for scalable and maintainable embedded system development.

Professional Experience

FullStak Developer, Infnova technologies 🔗 09/2025 – 02/2026
Adiss Ababa, Ethiopia

- Full-Stack Architecture: Developed high-performance systems using NestJS and Next.js, utilizing Prisma with PostgreSQL to automate complex cloud workflows and improve data reliability.
- Security Engineering: Eliminated 100% of credential leaks by routing Google Cloud requests through a secure Serverless Proxy layer protected by NestJS authentication guards.
- System Throughput: Built background processing pipelines using Redis, increasing simultaneous user tasks by 60% through optimized PostgreSQL connection pooling and query management.

FrontEnd Developer, Alle School of Fine Arts 🔗 09/2023 – 04/2024
Adiss Ababa, Ethiopia

- Developed a responsive showcase website for an art school using React.js, Tailwind CSS, and MongoDB.
- Optimized performance through reusable components and efficient frontend architecture for faster load times.
- Built scalable and user-friendly interfaces with seamless responsiveness across desktop and mobile devices.